全安全等级 Frodo 硬件实现评估报告

低资源开销设计（第二部分）

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版本记录

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1	V1.0	20250713	创建文档	赵正鹏、王家琛
2	V1.1	20250720	完成第一版报告	赵正鹏、王家琛
3	V1.2	20250725	增添 PTPX 分析与实际上板波形	赵正鹏、王家琛
4				
5				
6				
7				

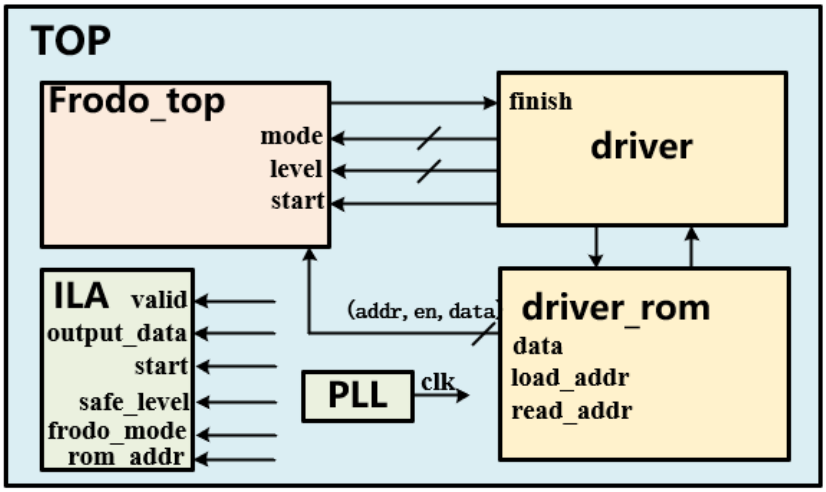
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1. 仿真结果

1.1. 仿真结果验证



上图为测试仿真方案，Frodo 本体（Frodo_top）与驱动（driver）被封装在一个 TOP 中，diver 控制 driver_rom 对 Frodo 内 RAM 进行数据的初始化，完成初始化后向本体发送安全等级（level）、具体算法（mode）、工作启动信号（start），开始执行 Frodo 算法，在一个算法流程执行完毕后，diver 控制 driver_rom 从 RAM 中读出运算结果数据。

输出数据的地址规划（用于结果的正确性验证）：

pkh(Keygen)	RAM1	0xFD2~0xFD5
ss(Encap)	RAM0	0xFFC~0xFFF
ss(Decap)	RAM1	0xFD6~0xFD9

仿真测试初始数据：

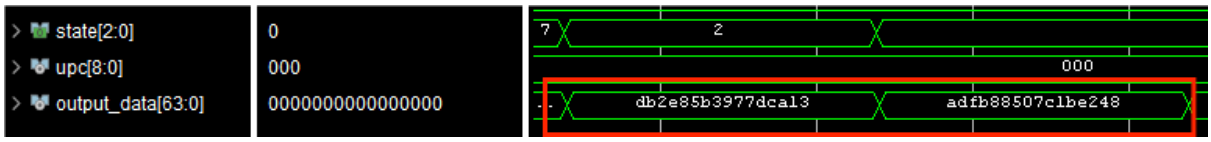
FrodoKEM 640 SHAKE	**s_seedSE_z randomness:** 7c9935a0b07694aa0c6d10e4db6b1add2fd81a25ccb148032dcd739936737f2d b505d7cfad1b497499323c8686325e4792f267aafa3f87ca60d01cb54f29202a **mu_salt randomness:** eb4a7c66ef4eba2ddb38c88d8bc706b1d639002198172a7b1942eca8f6c001ba2 6202bee59ac275484ea767d41d8d357
FrodoKEM 976 SHAKE	**s_seedSE_z randomness:** 7c9935a0b07694aa0c6d10e4db6b1add2fd81a25ccb148032dcd739936737f2d b505d7cfad1b497499323c8686325e4792f267aafa3f87ca60d01cb54f29202a3e 784ccb7ebcdcfd45542b7f6af778742e0f4479175084aa

	mu_salt randomness: ee716762c15e3b72aa7650a63b9a510040b03c0fe70475c0463bbc45a0ba5b7980dd46eef82fb062035077d042f306bb6391040e0dd965f1fda9d183ca9fccb48fc010b184ab0033
FrodoKEM 1344 SHAKE	**s_seedSE_z randomness:** 7c9935a0b07694aa0c6d10e4db6b1add2fd81a25ccb148032dcd739936737f2db505d7cfad1b497499323c8686325e4792f267aafa3f87ca60d01cb54f29202a3e784ccb7ebcdcfd45542b7f6af778742e0f4479175084aa488b3b74340678aa38e22e9628b0a161fdeb0bd252173b9c **mu_salt randomness:** 9f08587687ff66765c671de73e918d2823ca573ff4e7a31a9160324026e540eacb3a04e0d54c75deb9705bfdfbdf935a7528802ee6e5b0c6a73b2b761d9bd0848a6e4cf3fc4ca84f14e0331af35bfef41e42b13a6dae6df937f738c1857ba1ca

1.1.1. Frodo640

硬件仿真结果

Keygen 生成 pkh

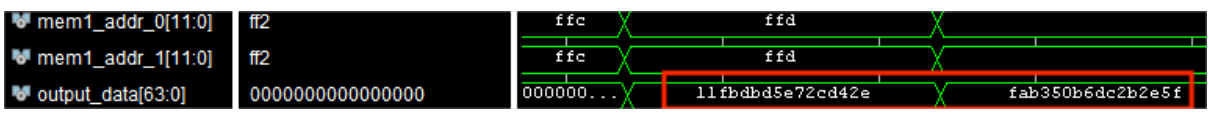


mem1_addr = fd2 , output_data = db2e85b3977dca13

mem1_addr = fd3 , output_data = adfb88507c1be248

软件结果(左为低位)： //pkh 13ca7d97b3852edb 48e21b7c5088fbad

Encaps 生成 ss



mem0_addr = ffc , output_data = 11fbdbd5e72cd42e

mem0_addr = ffd , output_data = fab350b6dc2b2e5f

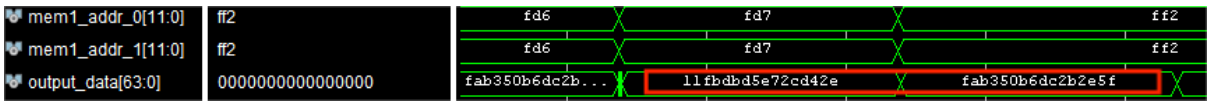
软件结果(左为低位)：

//ss(Encaps)

2ed42ce7d5dbfb11

5f2e2bdcb650b3fa

Decaps 生成 ss



mem1_addr = fd6 , output_data = 11fbdbd5e72cd42e

mem1_addr = fd7 , output_data = fab350b6dc2b2e5f

软件结果(左为低位)：

//ss(Decaps)

2ed42ce7d5dbfb11

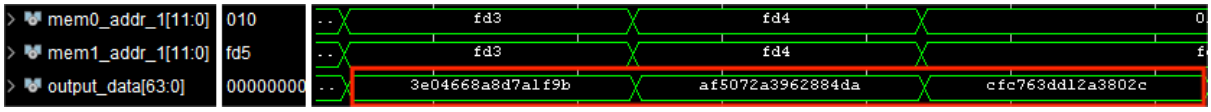
5f2e2bdcb650b3fa

通过比对，数据一致，说明 Frodo640 安全等级下，各算法流程执行正确

1.1.2. Frodo976

硬件仿真结果

Keygen 生成 pkh



mem1_addr = fd2 , output_data = 3e04668a8d7a1f9b

mem1_addr = fd3 , output_data = af5072a3962884da

mem1_addr = fd4 , output_data = cfc763dd12a3802c

软件结果(左为低位)：

//pkh

9b1f7a8d8a66043e

da842896a37250af

2c80a312dd63c7cf

Encaps 生成 ss



mem0_addr = ffc , output_data = e7f8d5a3695a6e5b

mem0_addr = ffd , output_data = d05e59956e3aea5e

mem0_addr = ffd , output_data = 4231378b5ba58d27

软件结果(左为低位)：

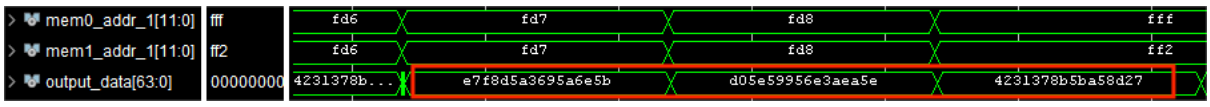
//ss(Encaps)

5b6e5a69a3d5f8e7

5eea3a6e95595ed0

278da55b8b373142

Decaps 生成 ss



mem1_addr = fd6 , output_data = e7f8d5a3695a6e5b

mem1_addr = fd7 , output_data = d05e59956e3aea5e

mem1_addr = fd8 , output_data = 4231378b5ba58d27

软件结果(左为低位)：

//ss(Dncaps)

5b6e5a69a3d5f8e7

5eea3a6e95595ed0

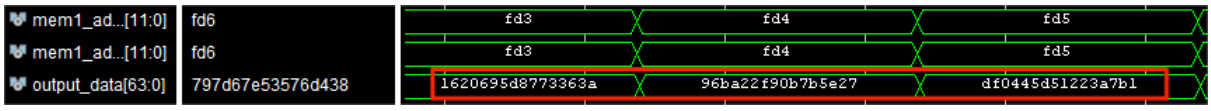
278da55b8b373142

通过比对，数据一致，说明 Frodo976 安全等级下，各算法流程执行正确

1.1.3. Frodo1344

硬件仿真结果

Keygen 生成 pkh



mem1_addr = fd2 , output_data = 1620695d8773363a

mem1_addr = fd3 , output_data = 96ba22f90b7b5e27

mem1_addr = fd4 , output_data = df0445d51223a7b1

mem1_addr = fd5 , output_data = 797d67e53576d438

软件结果(左为低位)：

//pkh

3a3673875d692016

275e7b0bf922ba96

b1a72312d54504df

38d47635e5677d79

Encaps 生成 ss



mem0_addr = ffc ,　output_data = 9ef14d4671f9208d

mem0_addr = ffd ,　output_data = e2d0af85d3bd6105

mem0_addr = ffd ,　output_data = 635ad4ef12e20cef

mem0_addr = ffe ,　output_data = ac6ad6f3642c5d2f

软件结果(左为低位)：

//ss(Encaps)

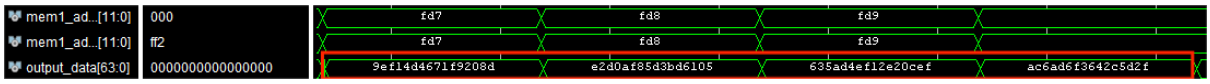
8d20f971464df19e

0561bdd385afd0e2

ef0ce212efd45a63

2f5d2c64f3d66aac

Decaps 生成 ss



mem1_addr = fd6 ,　output_data = 9ef14d4671f9208d

mem1_addr = fd7 ,　output_data = e2d0af85d3bd6105

mem1_addr = fd8 ,　output_data = 635ad4ef12e20cef

mem1_addr = fd9 ,　output_data = ac6ad6f3642c5d2f

软件结果(左为低位)：

//ss(Decaps)

8d20f971464df19e

0561bdd385afd0e2

ef0ce212efd45a63

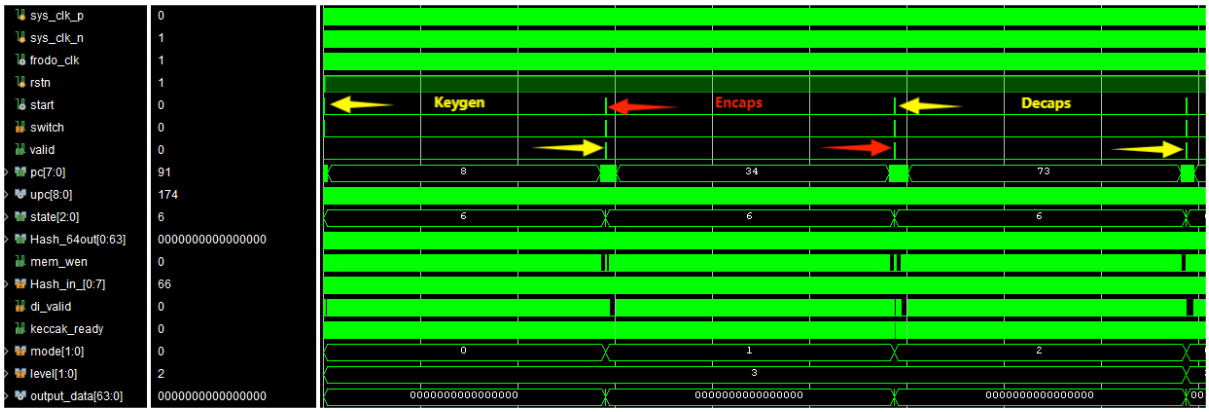
2f5d2c64f3d66aac

通过比对，数据一致，说明 Frodo1344 安全等级下，各算法流程执行正确

1.1.4.运算时间评估

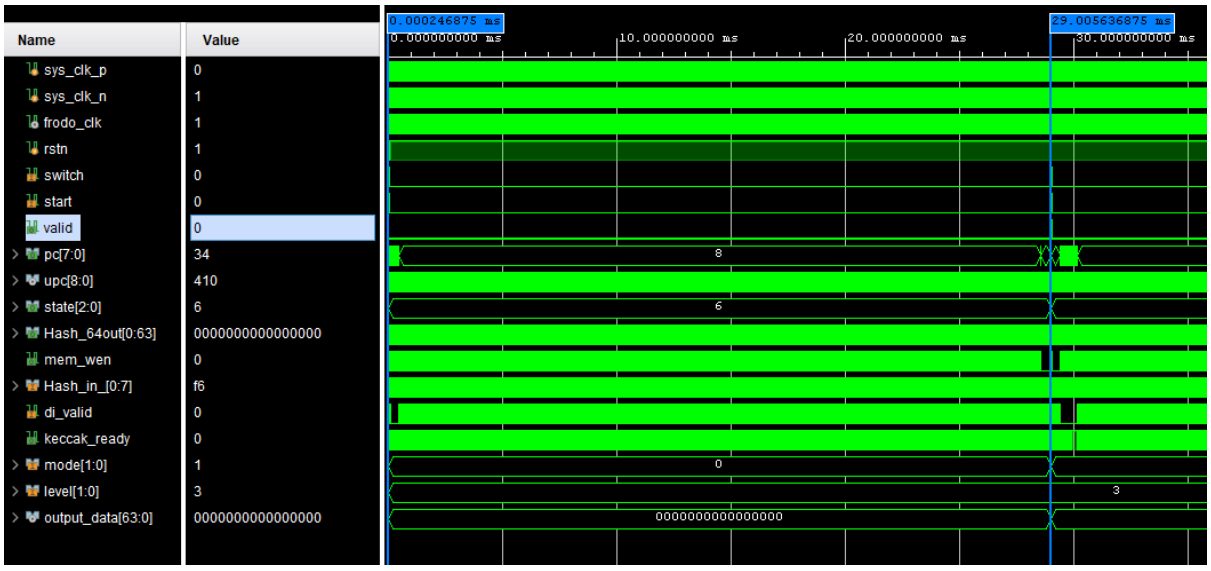
工程在 switch 信号上升沿来临开始进行某一算法流程测试，在初始化必要的输入数据后驱动会拉高 start 从而 Frodo 本体进行工作，运算结束后 valid 信号会拉高。一个算法流程的所耗时间从 start 信号的上升沿开始至 valid 信号上升沿结束。

以 Frodo640 为例，可见一个等级下的算法流程运行顺序



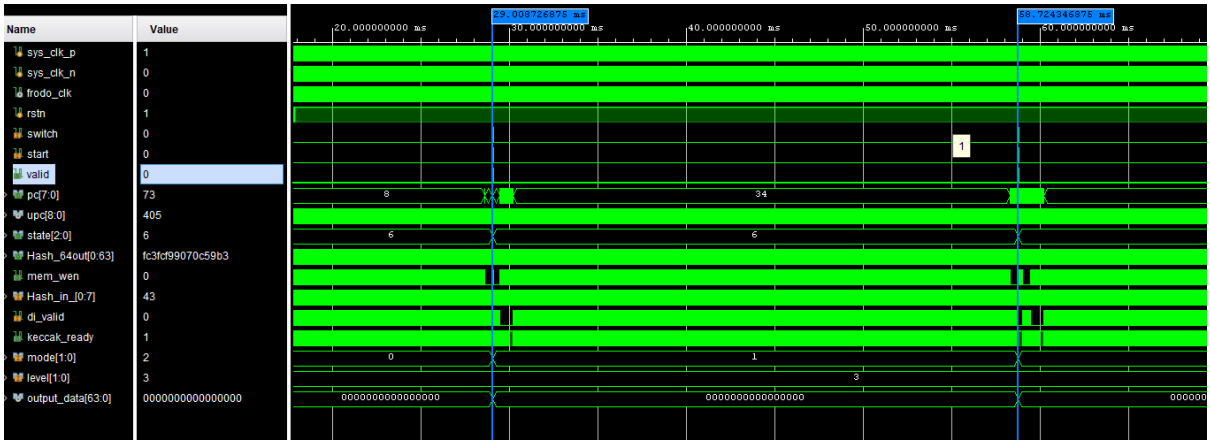
为方便计算，设置驱动时钟(Frodo_clk)为 100Mhz

Frodo640-keygen：



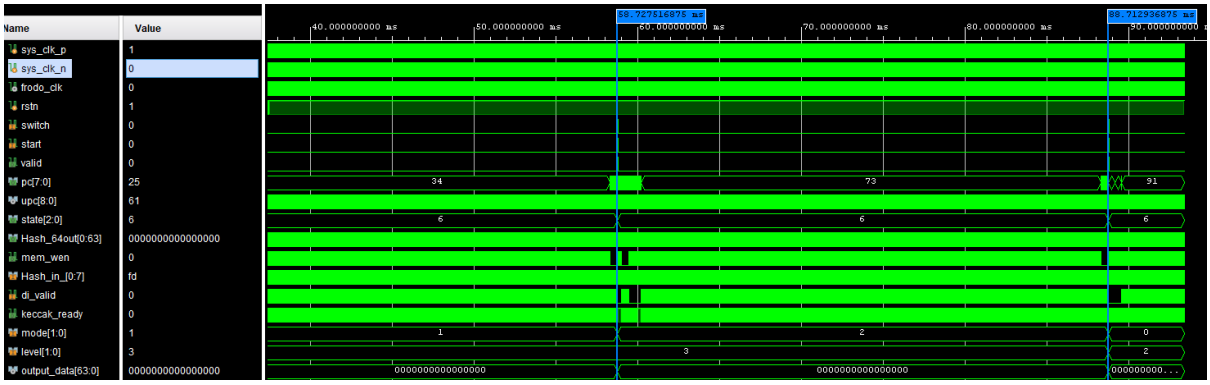
耗时 29,005,390ns，共计 2,900,539 个周期

Frodo640-Encaps：



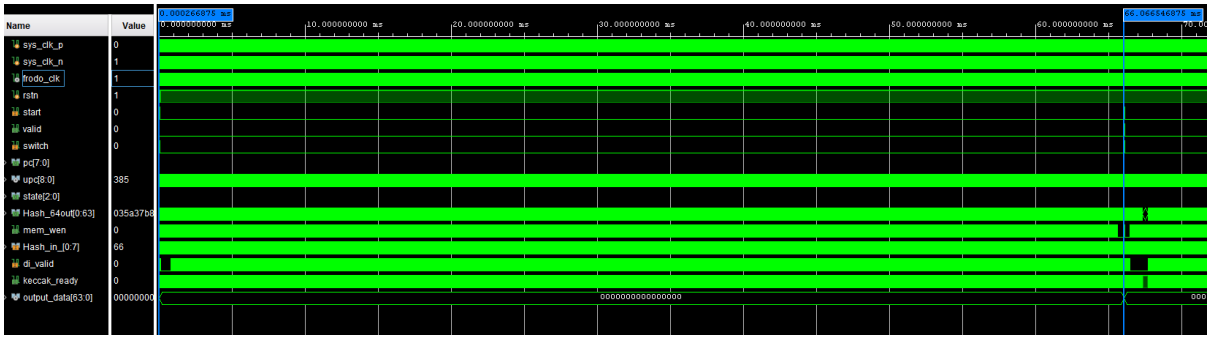
耗时 29,715,620ns，共计 2,971,562 个周期

Frodo640-Decaps：



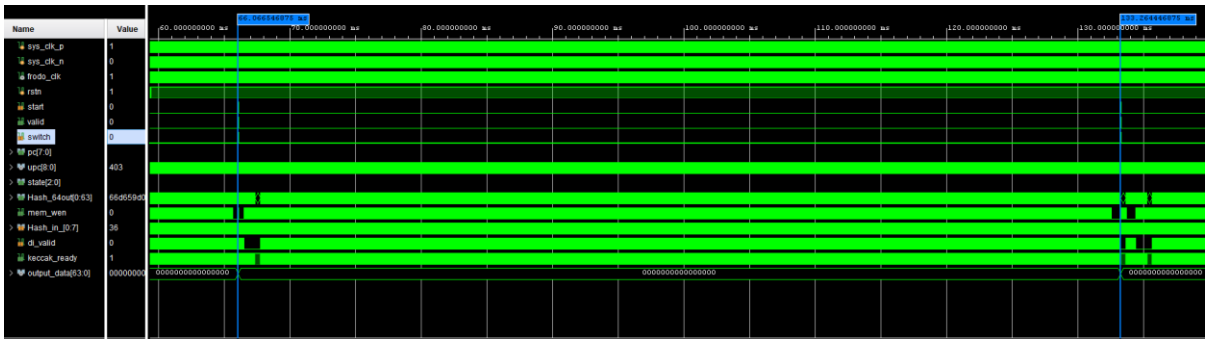
耗时 29,985,420ns，共计 2,998,542 个周期

Frodo976-keygen：



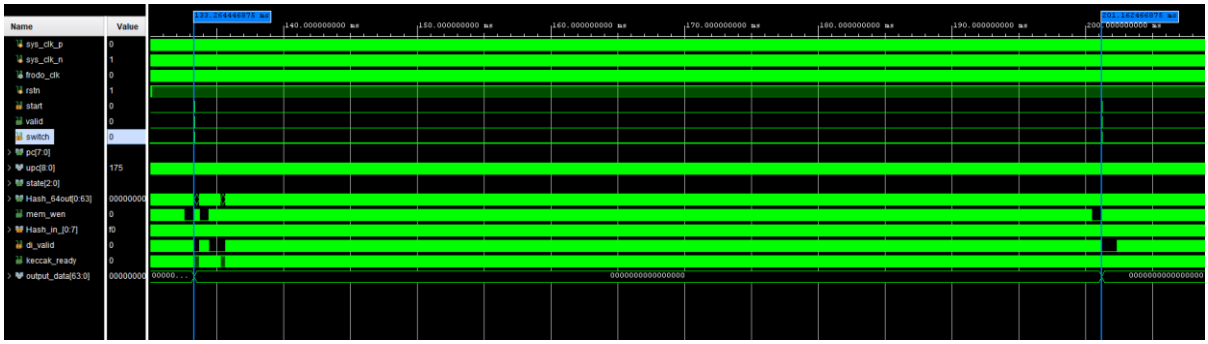
耗时 66,063,280ns，共计 6,606,328 个周期

Frodo976-Encaps：



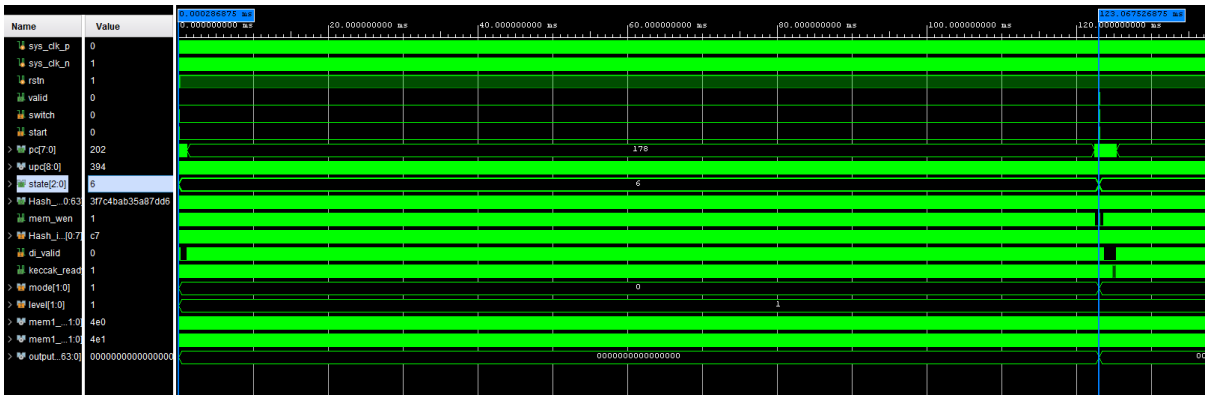
耗时 67,194,600ns，共计 6,719,460 个周期

Frodo976-Decaps



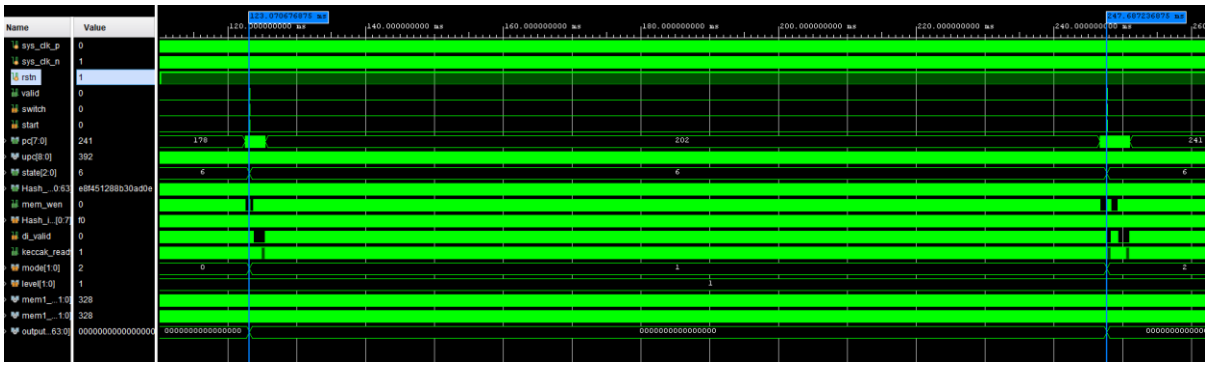
耗时 67,898,030ns，共计 6,789,803 个周期

Frodo1344-keygen



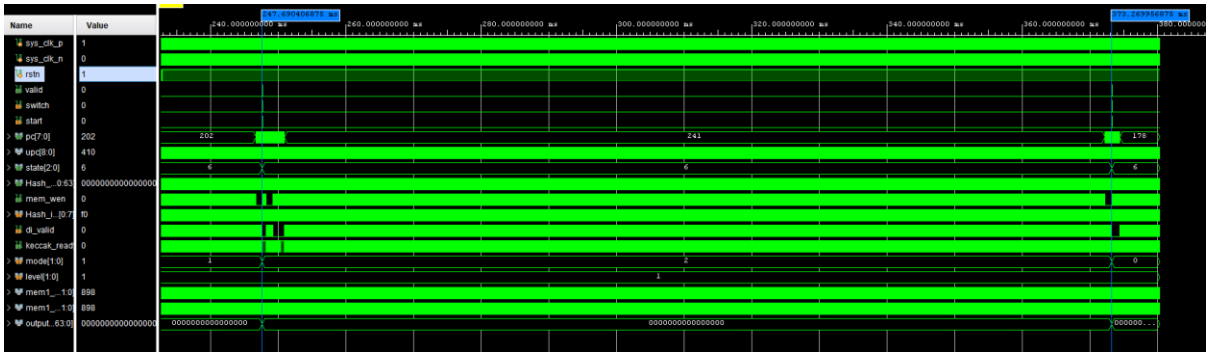
耗时 123,067,240ns，共计 12,306,724 个周期

Frodo1344-Encaps



耗时 124,616,560ns，共计 12,461,656 个周期

Frodo1344-Decaps



耗时 125,579,550ns，共计 12,557,955 个周期

运行周期数	Keygen	Encaps	Decaps
Frodo640	2,900,539	2,971,562	2,998,542
Frodo976	6,606,328	6,719,460	6,789,803
Frodo1344	12, 306, 724	12, 461, 656	12, 557, 955

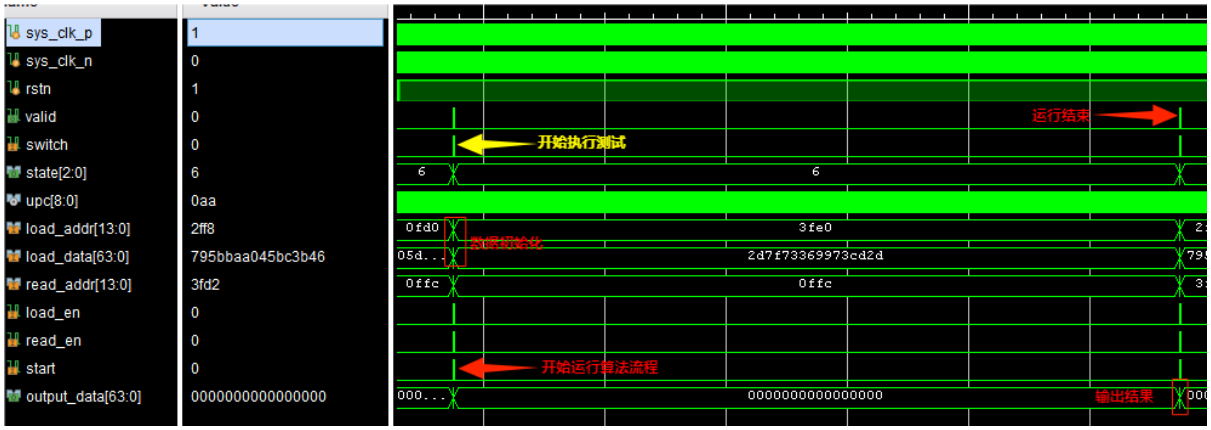
运行时间(100MHz 时钟下)	Keygen	Encaps	Decaps
Frodo640	29.01ms	29.71ms	29.99ms
Frodo976	66.06ms	67.19ms	67.90ms
Frodo1344	123.07ms	124.61ms	125.58ms

1.2. 子模块仿真波形

1.2.1. RAM 读写

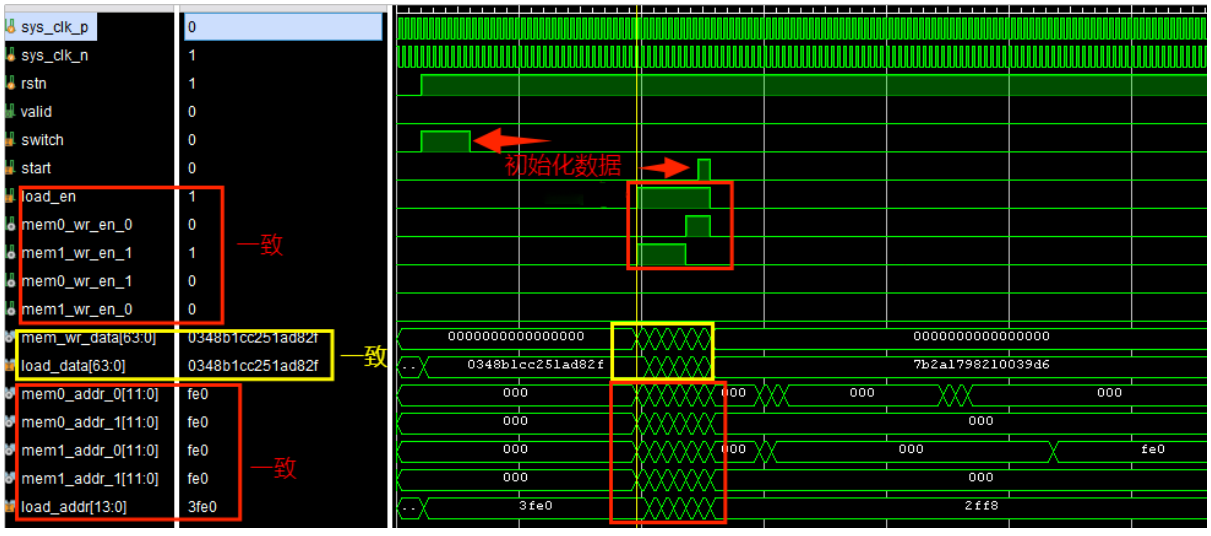
整体波形：

RAM 的直接相连端口为 mem_xx_xx 信号，通过驱动传入的 RAM 端口信号为 load_addr、load_data、load_en、read_addr、read_data，Frodo 本体控制的 RAM 端口信号为 Frodo_xx。在不同阶段 mem_xx 信号会与驱动传入信号或 Frodo 本体控制的 RAM 端口相连。



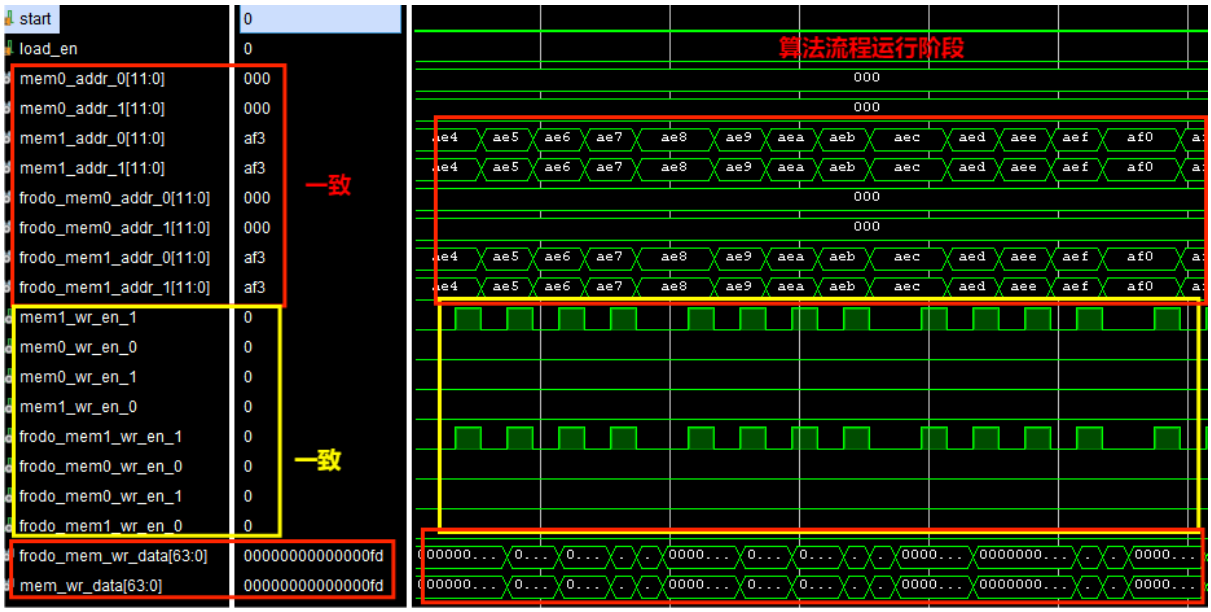
运行算法流程前的初始化数据：

此时 RAM 的地址与写数据端口分别受 load_addr 与 load_data 驱动，其中 load_addr 高两位为 RAM 索引区分此时控制 RAM1 还是 RAM0.



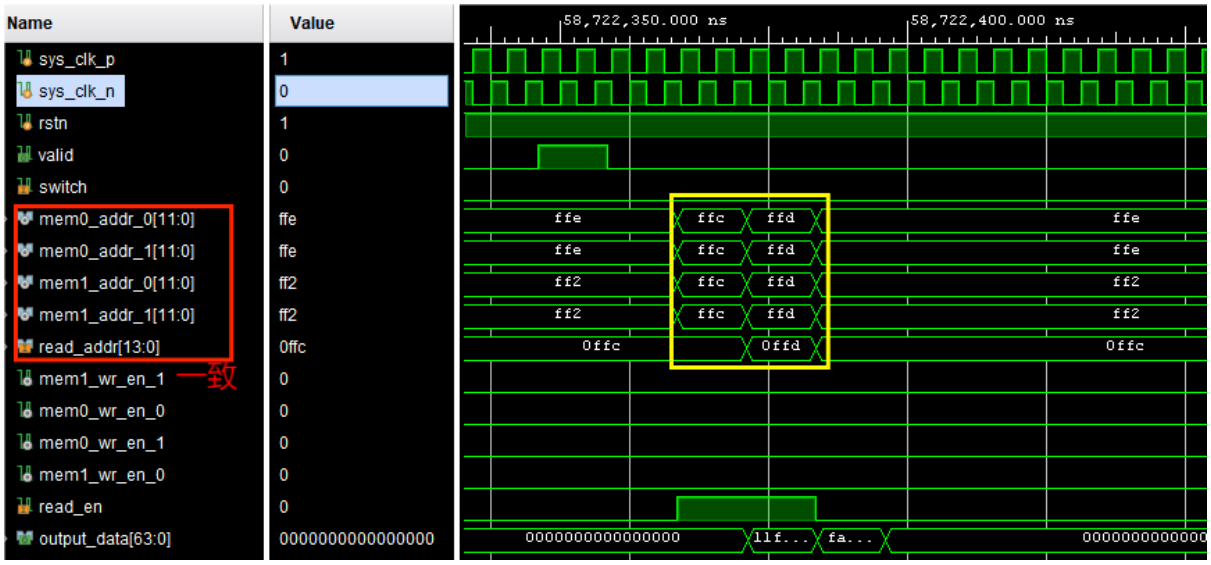
运行算法流程：

此时 RAM 的地址与写数据端口分别受 frodo_memx_addr_x 与 frodo_mem_wr_data 驱动。



结束算法流程后读出结果数据：

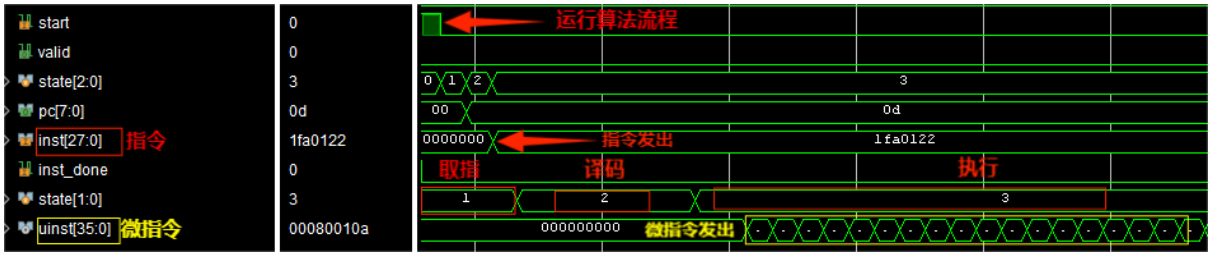
此时 RAM 的地址受 read_addr 驱动,读数据会连至 output_data。



1.2.2.指令执行

指令控制体系采取二级指令调度，一个指令指示一个工作(如:矩阵运算，输出 S)的运行，指令给出所需参数。通过待机-取指-译码-执行的流程控制子模块，在执行阶段，通过一条条微指令进行具体子模块信号的控制。

单指令下波形：



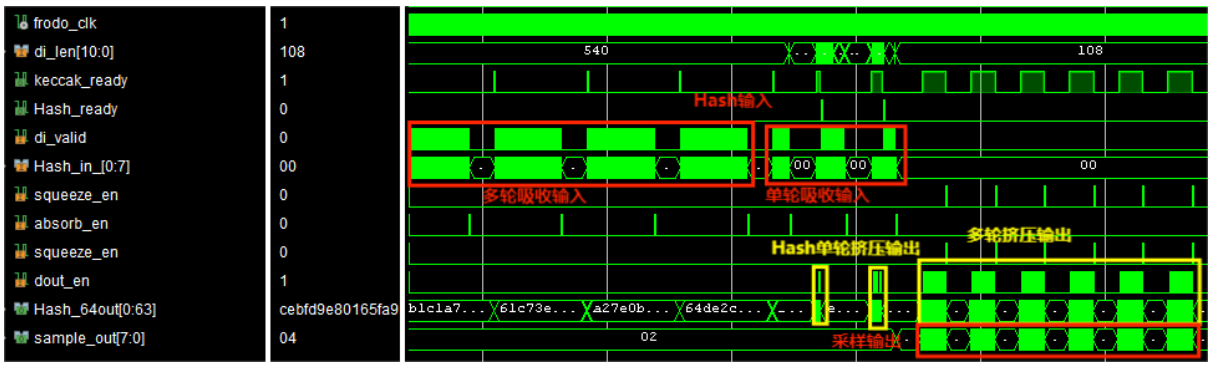
多个指令整体波形：



一个指令对应一个待机-取指-译码-执行的流程循环。

1.2.3. 哈希模块

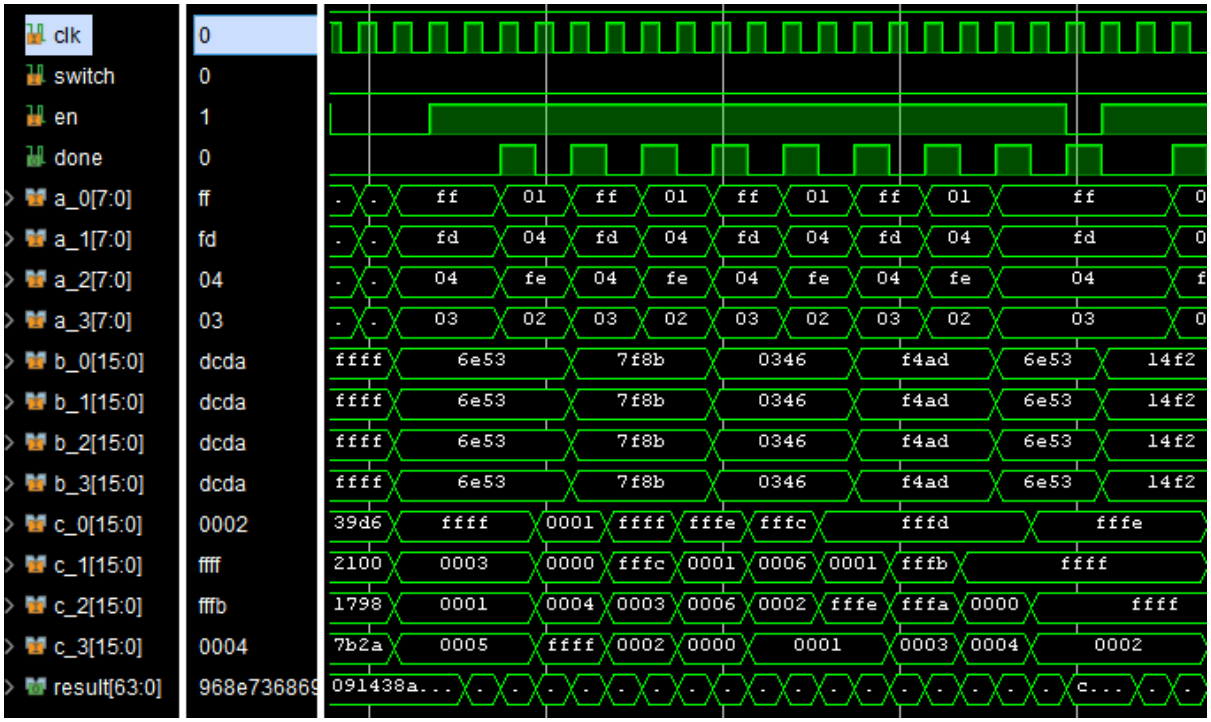
下图为 Hash 模块工作仿真波形：



挤压输出的结果直接连到采样输出，根据具体情况决定是直接采用挤压输出值还是采样输出值。

1.2.4. PE 阵列

下图 PE 阵列工作过程仿真波形：



输入数据与使能有效后，下一个周期输出乘积，再下一周期输出乘加运算结果。输出数据位宽为 64 位，即将四个计算结果并行输出。

2. 硬件评估结果

Vivado 版本:

Vivado v2024.2 (64-bit)
SW Build 5239630 on Fri Nov 08 22:35:27 MST 2024
IP Build 5239520 on Sun Nov 10 16:12:51 MST 2024
Tool Version Limit: 2024.11
Copyright 1986-2022 Xilinx, Inc. All Rights Reserved.
Copyright 2022-2024 Advanced Micro Devices, Inc. All Rights Reserved.
SharedData Build 5239561 on Fri Nov 08 14:39:27 MST 2024

FPGA 型号:xcku15p-ffve1517-2-i

2.1. Synthesis 结果

2.1.1.硬件资源评估

综合后硬件资源评估结果如下，共消耗 **6359** 个 LUT、**4428** 个 FF、**19.5** 个 BRAM 和 **4** 个 DSP。

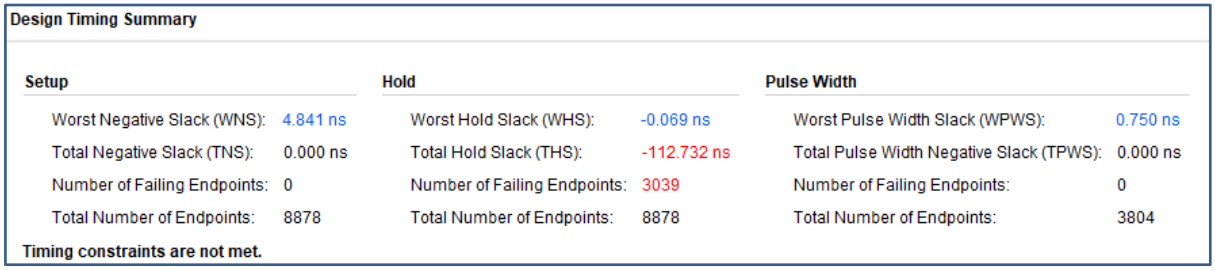
其中 Frodo 本体(u_Frodo_top)共消耗 **5205** 个 LUT、**2730** 个 FF、**16** 个 BRAM 和 **4** 个 DSP。

Name	CLB LUTs (522720)	CLB Registers (1045440)	CARRY8 (65340)	F7 Muxes (261360)	F8 Muxes (130680)	Block RAM Tile (984)	DSPs (1968)
Top	6359	4428	28	40	8	19.5	4
dbg_hub (dbg_hub_CV)	0	0	0	0	0	0	0
pll (clk_wiz_0)	0	0	0	0	0	0	0
u_driver (driver)	54	32	0	0	0	0	0
u_driver_rom (driver_rom)	62	18	0	5	0	1	0
u_Frodo_top (Frodo_top)	5205	2730	10	32	8	16	4
u_ila_0 (ila_0)	1031	1645	18	3	0	2.5	0

2.1.2.最高频率估计

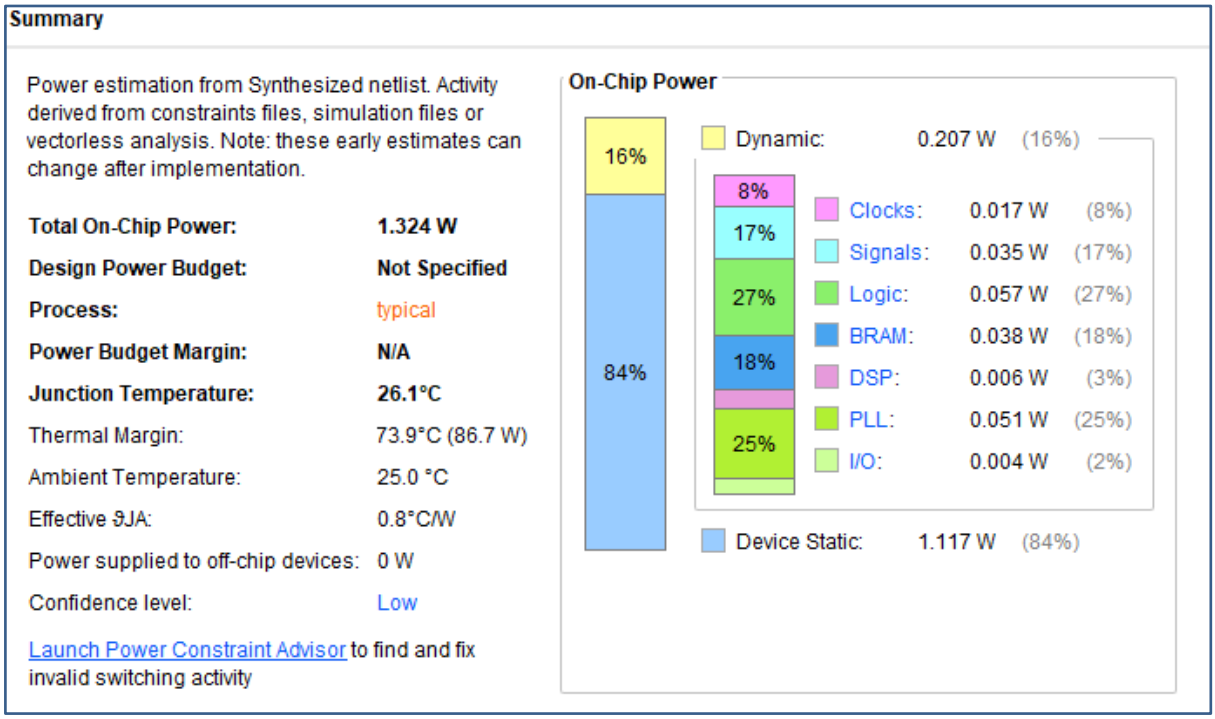
综合的时序评估如下，建立时间的裕度为 4.841ns，评估得到的最高频 193.8MHz。综合过程的 Hold Slack 不用考虑。

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock
Path 1	4.841	15	14	9	<hidden>	u_Frodo_top/u_...sult_reg[15]/D	4.972	3.249	1.723	10.0	clk_out1_clk_wiz_0	clk_out1_clk_wiz_0
Path 2	4.841	15	14	9	<hidden>	u_Frodo_top/u_...sult_reg[15]/D	4.972	3.249	1.723	10.0	clk_out1_clk_wiz_0	clk_out1_clk_wiz_0
Path 3	4.841	15	14	9	<hidden>	u_Frodo_top/u_...sult_reg[15]/D	4.972	3.249	1.723	10.0	clk_out1_clk_wiz_0	clk_out1_clk_wiz_0
Path 4	4.843	15	14	9	<hidden>	u_Frodo_top/u_...sult_reg[13]/D	4.970	3.248	1.722	10.0	clk_out1_clk_wiz_0	clk_out1_clk_wiz_0
Path 5	4.843	15	14	9	<hidden>	u_Frodo_top/u_...sult_reg[13]/D	4.970	3.248	1.722	10.0	clk_out1_clk_wiz_0	clk_out1_clk_wiz_0
Path 6	4.843	15	14	9	<hidden>	u_Frodo_top/u_...sult_reg[13]/D	4.970	3.248	1.722	10.0	clk_out1_clk_wiz_0	clk_out1_clk_wiz_0
Path 7	4.856	15	14	9	<hidden>	u_Frodo_top/u_...sult_reg[14]/D	4.957	3.234	1.723	10.0	clk_out1_clk_wiz_0	clk_out1_clk_wiz_0
Path 8	4.856	15	14	9	<hidden>	u_Frodo_top/u_...sult_reg[14]/D	4.957	3.234	1.723	10.0	clk_out1_clk_wiz_0	clk_out1_clk_wiz_0
Path 9	4.856	15	14	9	<hidden>	u_Frodo_top/u_...sult_reg[14]/D	4.957	3.234	1.723	10.0	clk_out1_clk_wiz_0	clk_out1_clk_wiz_0
Path 10	4.877	15	14	9	<hidden>	u_Frodo_top/u_...sult_reg[12]/D	4.936	3.214	1.722	10.0	clk_out1_clk_wiz_0	clk_out1_clk_wiz_0



2.1.3. 功耗评估

功耗评估如下：



2.2. Implementation 结果

2.2.1. 硬件资源评估

实现后硬件资源评估结果如下，共消耗 **6640** 个 LUT、**4923** 个 FF、**19.5** 个 BRAM 和 **4** 个 DSP。

其中 Frodo 本体(u_Frodo_top)共消耗 **5144** 个 LUT、**2530** 个 FF、**16** 个 BRAM 和 **4** 个 DSP。

Name	CLB LUTs (522720)	CLB Registers (1045440)	CARRY8 (65340)	F7 Muxes (261360)	F8 Muxes (130680)	CLB (65340)	LUT as Logic (522720)	LUT as Memory (161280)	Block RAM Tile (984)	DSPs (1968)
Top	6640	4923	35	40	8	1304	6261	379	19.5	4
dbg_hub (dbg_hub)	428	741	7	0	0	123	408	20	0	0
pll (clk_wiz_0)	0	0	0	0	0	0	0	0	0	0
u_driver (driver)	52	32	0	0	0	13	52	0	0	0
u_driver_rom (driver_rom)	61	18	0	5	0	35	61	0	1	0
u_Frodo_top (Frodo_top)	5144	2530	10	32	8	914	4946	198	16	4
u_ila_0 (ila_0)	951	1599	18	3	0	270	790	161	2.5	0

2.2.2.最高频率估计

实现后的时序评估如下，建立时间的裕度为 2.682ns，评估得到的最高频 136.6MHz。

Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exc...	Clock Uncertainty
Path 21	15.273	3	50	dbg_hub/inst/BS.../state_reg[0]/C	dbg_hub/inst/B...bscan_inst/TDO	1.797	0.384	1.413	25.0	dbg_hub/inst/BSCANID.u_xsdbm_i	dbg_hub/inst/...INTERNAL_TCK		0.235
Path 22	20.489	1	79	dbg_hub/inst/...INTERNAL_TCK	dbg_hub/inst/BS...ftreg_reg[12]/R	6.525	5.122	1.403	25.0	dbg_hub/inst/BSCANID.u_xsdbm_i	dbg_hub/inst/...INTERNAL_TCK		0.235
Path 23	20.489	1	79	dbg_hub/inst/...INTERNAL_TCK	dbg_hub/inst/BS...ftreg_reg[13]/R	6.525	5.122	1.403	25.0	dbg_hub/inst/BSCANID.u_xsdbm_i	dbg_hub/inst/...INTERNAL_TCK		0.235
Path 24	20.489	1	79	dbg_hub/inst/...INTERNAL_TCK	dbg_hub/inst/BS...ftreg_reg[14]/R	6.525	5.122	1.403	25.0	dbg_hub/inst/BSCANID.u_xsdbm_i	dbg_hub/inst/...INTERNAL_TCK		0.235
Path 25	20.489	1	79	dbg_hub/inst/...INTERNAL_TCK	dbg_hub/inst/BS...ftreg_reg[15]/R	6.525	5.122	1.403	25.0	dbg_hub/inst/BSCANID.u_xsdbm_i	dbg_hub/inst/...INTERNAL_TCK		0.235
Path 26	20.492	1	79	dbg_hub/inst/...INTERNAL_TCK	dbg_hub/inst/BS...ftreg_reg[16]/R	6.525	5.122	1.403	25.0	dbg_hub/inst/BSCANID.u_xsdbm_i	dbg_hub/inst/...INTERNAL_TCK		0.235
Path 27	20.492	1	79	dbg_hub/inst/...INTERNAL_TCK	dbg_hub/inst/BS...ftreg_reg[17]/R	6.525	5.122	1.403	25.0	dbg_hub/inst/BSCANID.u_xsdbm_i	dbg_hub/inst/...INTERNAL_TCK		0.235
Path 28	20.492	1	79	dbg_hub/inst/...INTERNAL_TCK	dbg_hub/inst/BS...ftreg_reg[18]/R	6.525	5.122	1.403	25.0	dbg_hub/inst/BSCANID.u_xsdbm_i	dbg_hub/inst/...INTERNAL_TCK		0.235
Path 29	20.492	1	79	dbg_hub/inst/...INTERNAL_TCK	dbg_hub/inst/BS...ftreg_reg[22]/R	6.525	5.122	1.403	25.0	dbg_hub/inst/BSCANID.u_xsdbm_i	dbg_hub/inst/...INTERNAL_TCK		0.235
Path 30	20.700	2	35	dbg_hub/inst/...INTERNAL_TCK	dbg_hub/inst/BS...curid_reg[31]/D	6.386	5.213	1.173	25.0	dbg_hub/inst/BSCANID.u_xsdbm_i	dbg_hub/inst/...INTERNAL_TCK		0.235

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	2.682 ns	Worst Hold Slack (WHS):	0.007 ns	Worst Pulse Width Slack (WPWS):	0.750 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	12407	Total Number of Endpoints:	12391	Total Number of Endpoints:	5684

All user specified timing constraints are met.

2.2.3.功耗评估

功耗评估如下：

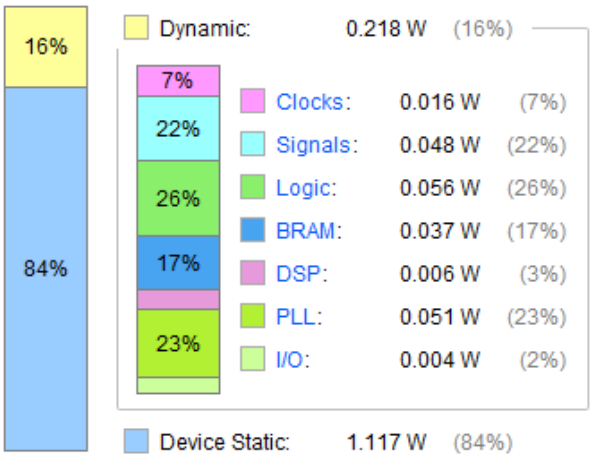
Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	**1.335 W**
Design Power Budget:	**Not Specified**
Process:	typical
Power Budget Margin:	**N/A**
Junction Temperature:	**26.1°C**
Thermal Margin:	73.9°C (86.7 W)
Ambient Temperature:	25.0 °C
Effective ϑJA:	0.8°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	Medium

Launch Power Constraint Advisor to find and fix invalid switching activity

On-Chip Power



2.3. 等效逻辑门数评估

以下为 DC 综合所得 Total cell area

```
Combinational area:              14102.591884
Buf/Inv area:                      694.847991
Noncombinational area:           17709.551593
Macro/Black Box area:           134908.872314
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                166721.015792
Total area:                 undefined
```

可得 Total cell area 为 166721.015792，保守估计等效逻辑门数为 315,043.49，在 31 万等效逻辑门左右，远小于预期的 100 万门，且在不同驱动能力下等效逻辑门数可能有所降低，符合小于 100 万的门要求。

2.4. PTPX 功耗分析

2.4.1. 在 16Mhz 时钟下功耗分析

16Mhz 时钟下的 PTPX 功耗分析报告如下：

```
                                   Int      Switch   Leak     Total
Hierarchy                          Power    Power    Power    Power    %
-------------------------------------------------------------------------------
Frodo_top                          1.28e-03 1.38e-05 2.49e-02 2.62e-02 100.0
  u_keccak (keccak)                2.71e-04 7.79e-06 1.76e-02 1.78e-02  68.2
    u_hash_out_buffer (hash_out_buffer)
                                   7.69e-05 8.24e-07 4.16e-03 4.24e-03  16.2
  u_control (Control)              8.13e-05 2.16e-06 2.28e-03 2.37e-03   9.1


                                   Peak     Peak           Glitch   X-tran
Hierarchy                          Power    Time           Power    Power
-------------------------------------------------------------------------------
Frodo_top                            8.069 2795406.249-2795406.250
                                                                0.000 1.15e-10
  u_keccak (keccak)                  0.508 1874093.750-1874093.751
                                                                0.000 1.46e-11
    u_hash_out_buffer (hash_out_buffer)
                                     0.432 1874093.750-1874093.751
                                                                0.000    0.000
  u_control (Control)                2.950 3008906.249-3008906.250
                                                                0.000 9.51e-11
```

由图可知 16Mhz 下 PTPX 分析出的总峰值功耗为 8.069w，其他数据如下表所示：

模块	Int Power(w)	Swtich Power(w)	Leak Power(w)	Total Power(w)
总体工程	$1.28e^{-3}$	$1.38e^{-5}$	$2.49e^{-2}$	$2.62e^{-2}$
keccak	$2.71e^{-4}$	$7.79e^{-6}$	$1.76e^{-2}$	$1.78e^{-2}$
Control	$8.13e^{-5}$	$2.16e^{-6}$	$2.28e^{-3}$	$2.37e^{-3}$
其他模块	$9.28e^{-4}$	$3.85e^{-6}$	$5.02e^{-3}$	$6.03e^{-3}$

2.4.2.在 24Mhz 时钟下功耗分析

16Mhz 时钟下的 PTPX 功耗分析报告如下：

```
                                   Int       Switch    Leak      Total
Hierarchy                          Power     Power     Power     Power     %
------------------------------------------------------------------------------
Frodo_top                          1.91e-03  2.07e-05  2.49e-02  2.68e-02  100.0
  u_keccak (keccak)                4.07e-04  1.17e-05  1.76e-02  1.80e-02   67.1
    u_hash_out_buffer (hash_out_buffer)
                                   1.15e-04  1.24e-06  4.16e-03  4.27e-03   15.9
  u_control (Control)              1.22e-04  3.24e-06  2.28e-03  2.41e-03    9.0


                                   Peak      Peak                Glitch    X-tran
Hierarchy                          Power     Time                Power     Power
------------------------------------------------------------------------------
Frodo_top                             8.069  1863574.349-1863574.350
                                                                 0.000  1.73e-10
  u_keccak (keccak)                   0.508  1249375.842-1249375.843
                                                                 0.000  2.18e-11
    u_hash_out_buffer (hash_out_buffer)
                                      0.432  1249375.842-1249375.843
                                                                 0.000     0.000
  u_control (Control)                 2.950  2005905.405-2005905.406
                                                                 0.000  1.43e-10
```

由图可知 24Mhz 下 PTPX 分析出的总峰值功耗为 8.069w，其他数据如下表所示：

模块	Int Power(w)	Swtich Power(w)	Leak Power(w)	Total Power(w)
总体工程	$1.91e^{-3}$	$2.07e^{-5}$	$2.49e^{-2}$	$2.68e^{-2}$
keccak	$4.07e^{-4}$	$1.17e^{-5}$	$1.76e^{-2}$	$1.80e^{-2}$
Control	$1.22e^{-4}$	$3.24e^{-6}$	$2.28e^{-3}$	$2.41e^{-3}$
其他模块	$1.38e^{-3}$	$5.76e^{-6}$	$5.02e^{-3}$	$6.39e^{-3}$

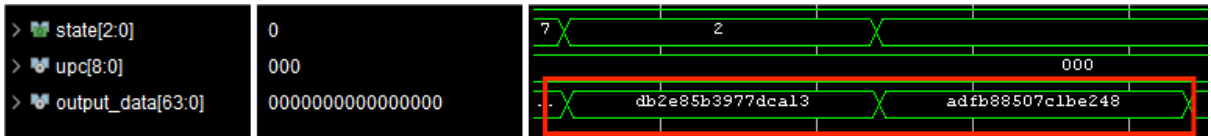
3. 上板验证结果

以下结果系统时钟为 200MHz，Frodo_top 的时钟频率为 pll 的输出频率为 100MHz。

3.1. Frodo640

3.1.1. Keygen 生成 pkh

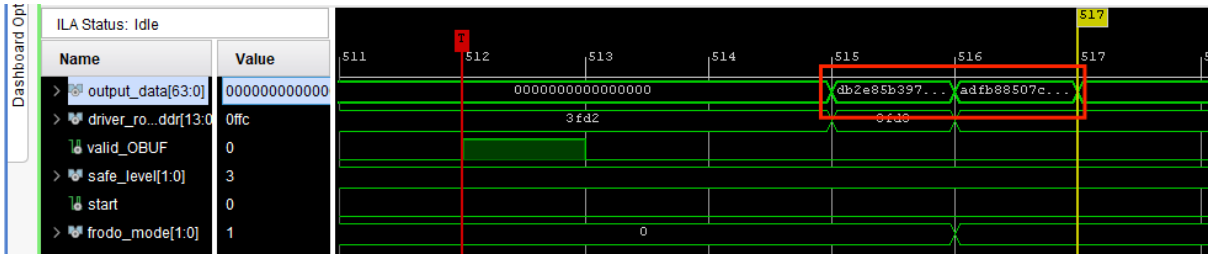
硬件仿真结果：



mem1_addr = fd2 ,　output_data = db2e85b3977dca13

mem1_addr = fd3 ,　output_data = adfb88507c1be248

上板波形：



3.1.2. Encaps 生成 ss

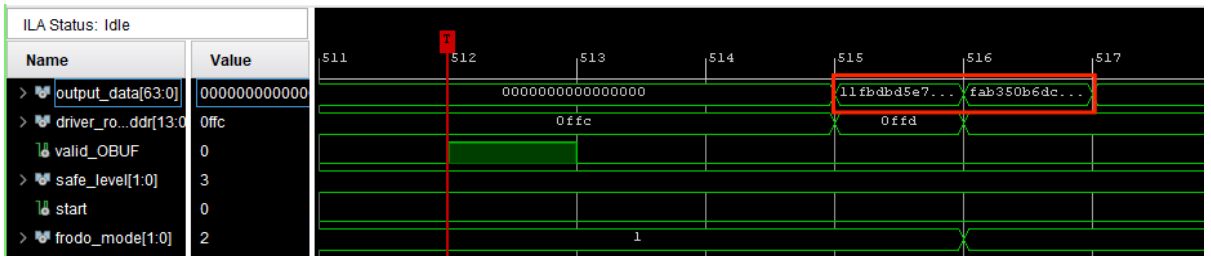
硬件仿真结果：



mem0_addr = ffc ,　output_data = 11fbdbd5e72cd42e

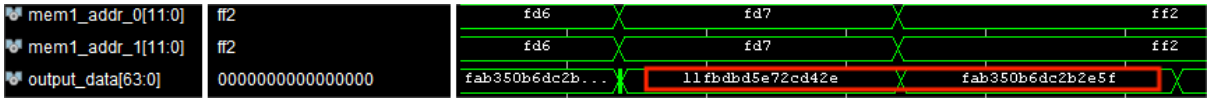
mem0_addr = ffd ,　output_data = fab350b6dc2b2e5f

上板波形：



3.1.3. Decaps 生成 ss

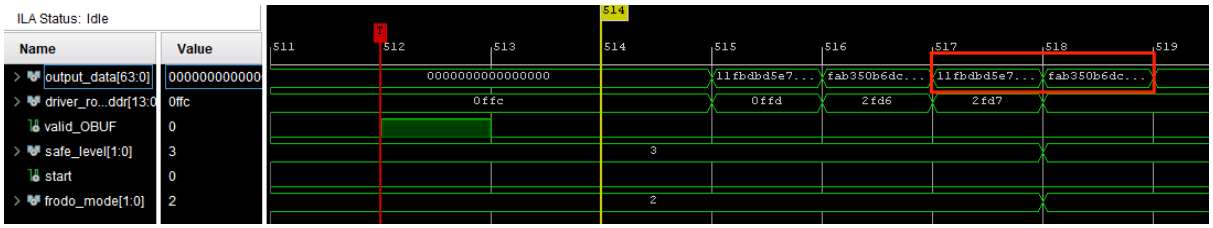
硬件仿真结果：



mem1_addr = fd6 ， output_data = 11fbdbd5e72cd42e

mem1_addr = fd7 ， output_data = fab350b6dc2b2e5f

上板波形：

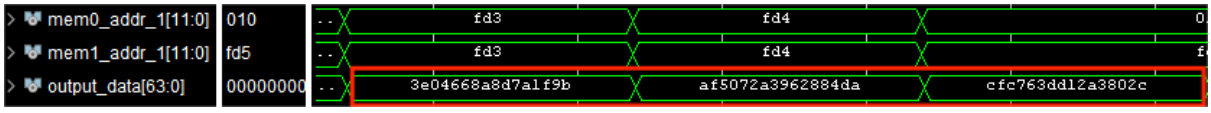


通过比对，数据一致，说明 Frodo640 安全等级下，各算法流程执行正确

3.2. Frodo976

3.2.1. Keygen 生成 pkh

硬件仿真结果：



mem1_addr = fd2 ， output_data = 3e04668a8d7a1f9b

mem1_addr = fd3 ， output_data = af5072a3962884da

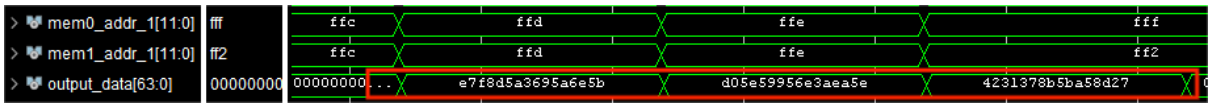
mem1_addr = fd4 ， output_data = cfc763dd12a3802c

上板波形：



3.2.2. Encaps 生成 ss

硬件仿真结果：

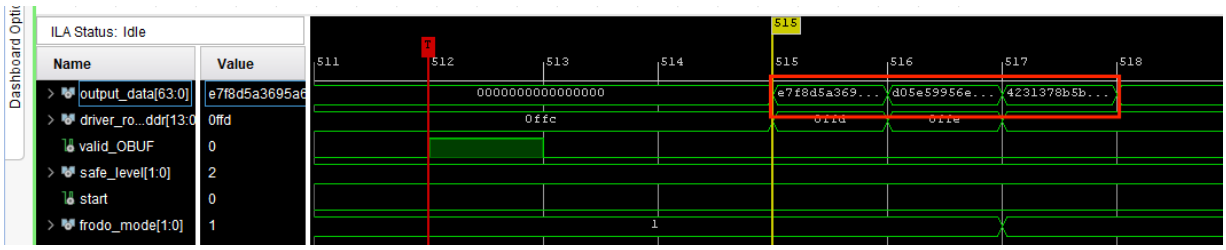


mem0_addr = ffc ， output_data = e7f8d5a3695a6e5b

mem0_addr = ffd ， output_data = d05e59956e3aea5e

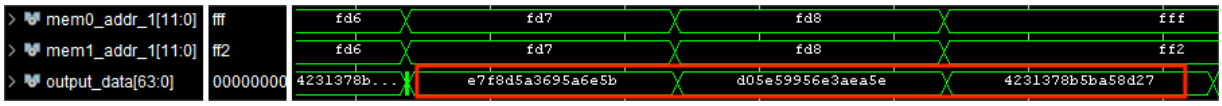
mem0_addr = ffd ， output_data = 4231378b5ba58d27

上板波形：



3.2.3. Decaps 生成 ss

硬件仿真结果：

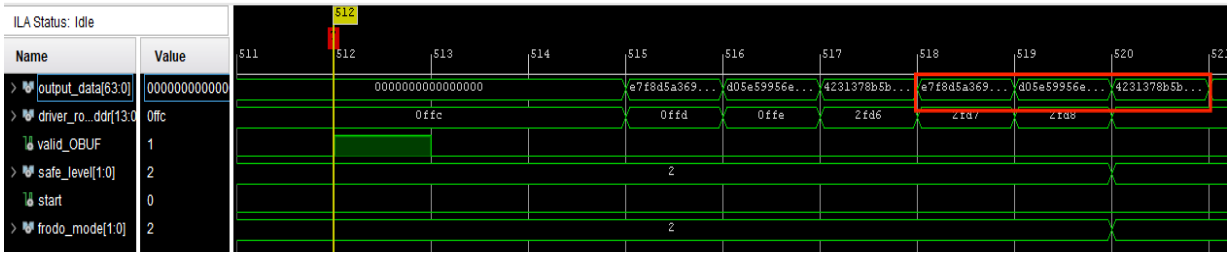


mem1_addr = fd6 ， output_data = e7f8d5a3695a6e5b

mem1_addr = fd7 ， output_data = d05e59956e3aea5e

mem1_addr = fd8 ， output_data = 4231378b5ba58d27

上板波形：

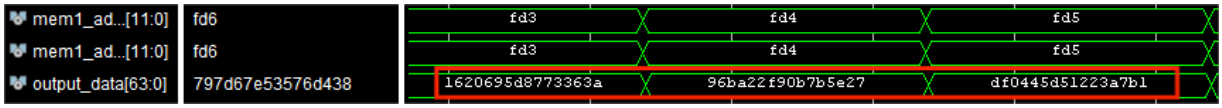


通过比对，数据一致，说明 Frodo976 安全等级下，各算法流程执行正确

3.3. Frodo1344

3.3.1. Keygen 生成 pkh

硬件仿真结果：



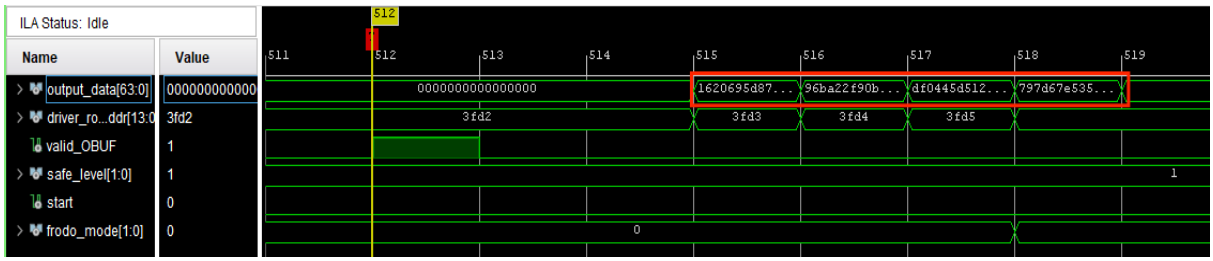
mem1_addr = fd2 ， output_data = 1620695d8773363a

mem1_addr = fd3 ， output_data = 96ba22f90b7b5e27

mem1_addr = fd4 ， output_data = df0445d51223a7b1

mem1_addr = fd5 ， output_data = 797d67e53576d438

上板波形：



3.3.2. Encaps 生成 ss

硬件仿真结果：



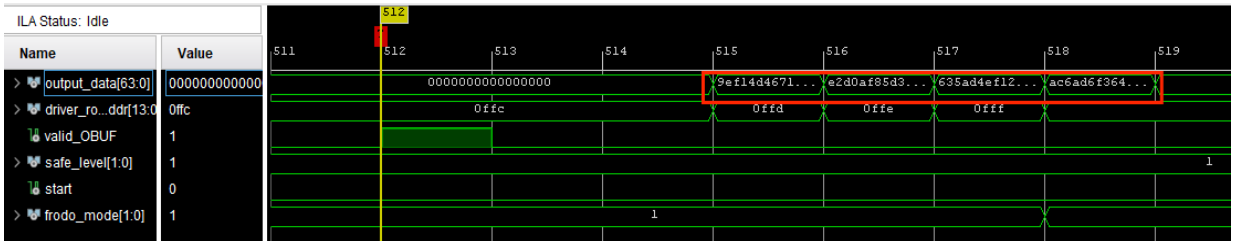
mem0_addr = ffc ,　output_data = 9ef14d4671f9208d

mem0_addr = ffd ,　output_data = e2d0af85d3bd6105

mem0_addr = ffd ,　output_data = 635ad4ef12e20cef

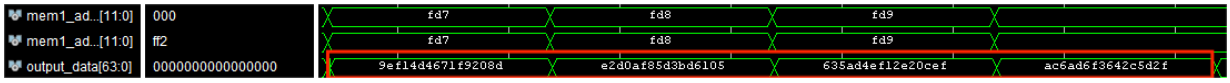
mem0_addr = ffe ,　output_data = ac6ad6f3642c5d2f

上板波形：



3.3.3. Decaps 生成 ss

硬件仿真结果：



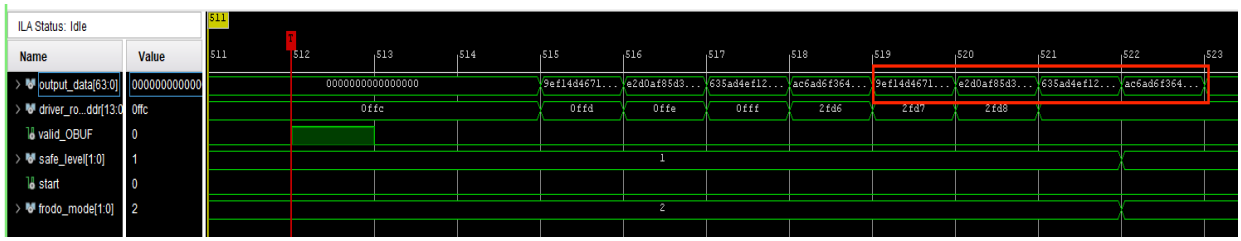
mem1_addr = fd6 ,　output_data = 9ef14d4671f9208d

mem1_addr = fd7 ,　output_data = e2d0af85d3bd6105

mem1_addr = fd8 ,　output_data = 635ad4ef12e20cef

mem1_addr = fd9 ,　output_data = ac6ad6f3642c5d2f

上板波形：



通过比对，数据一致，说明 Frodo1344 安全等级下，各算法流程执行正确

4. 附录

4.1. 算法执行指令

4.1.1. Frodo640

Keygen
11001011111000000000000000011 //吸收 5f　absorb_s
00011111110000000001000100000 //吸收 seedse absorb 长度 256
01110101001000000000000100100 // genS
01100000001000000000000000101 //genE
00011111111000000000100000010 //吸收 z　absorb 长度 128
01000000000000000000100000001 //absorb_ex,长度 128
01011111101000000000100100000 //生成 seedA,长度 128
11100010011001000000000000000 //Amatmul B = AS + E
00011111101000000000100100010 //absorb 吸收 seedA,长度 128
00000000001000000000000000001 // pkh,吸收矩阵 B
01000000000000000001010000001 //执行一次吸收
01011111101001000000100100000 //挤出 pkh,长度 128
Encaps
00011111101000000000100100010
00000000001000000000000000001
01000000000000000001010000001
01011111101001000000100100000　//生成 pkh
00011111101001000000100100010　//吸收 pkh
00011111111001000000100000000　//吸收 u
00011111111100000001000100000　//吸收 salt
01000000000000000010000000001　//执行吸收，长度 512
01011111110000000001000100000　//生成 seedse，长度 256
01011111111011000000100000000　//生成 k，长度 128
11010010110000000000000000011 //吸收 96
00011111110000000001000100000 //吸收 seedse
01110101001000000000000000100 //生成 S'
01100000001000000000000101101 //生成 E'
01011111101000000000100000100 //生成 E"前 1 行（128bit）
10100000000000000000000000001 //执行一次挤压

0101111110100100011100000100 //生成 E"后 7 行 （896bit）
1010100010001010000000000000 //转置 E"
1000110001001010000000000000 // 生成 V_T
1010101010101000000000000000 //转置 V
1100111011101010000000000000 //encodeu
1110110100110000000000000001 // B'=S'A+E'
1000100010001010000000000001 //C = V+encode(u)
0010000000100000000000100000 // 吸收 B'前半段
0010000000100010000000100001 //吸收 B'后半段
0001111111010000100001110000 //吸收 C
0001111111110000000110100000 //吸收 salt 前 192bit
0100000000000000101010000001 //执行一次吸收
0001111111110110000010100000 //吸收 salt 后 64bit
0001111111101100000100000000 //吸收 k 128bit
0100000000000000000110000001 //执行一次吸收
0101111111111000000100000000 //挤出 ss,128
Decaps
1001011100001001000000000000 //M=C-B'S
1100100010010101000000000000 // u' = decode(u)
0001111110100100000100100010 //吸收 pkh
0001111111100100000100000000 //吸收 u
0001111111101100001000100000 //吸收 salt
0100000000000000010000000001 //执行吸收
0101111111000000001000100000 //生成 seedse'
0101111111101100000100000000 //生成 k'
0010000000100000000000100000 // 吸收 B'前半段
0010000000100010000000100001 //吸收 B'后半段
0001111110100000000001110000 //吸收 C
0001111111110000000110100000 //吸收 salt 前 192bit
0100000000000000101010000001 //执行一次吸收
0001111111110110000010100000 //吸收 salt 后 64bit
0001111111101100000100000000 //吸收 k 128bit
0100000000000000000110000001 //执行一次吸收
0101111110101100100100100000 //挤出 ss'
1101001011000000000000000011 //吸收 96

00011111110000000010001000000 //吸收 seedse
01110101001000000000000000100 //生成 S'
01100000001000000000000101101 //生成 E'
01011111101000000001000000100 //生成 E"前两行
10100000000000000000000000001 //执行一次挤压
01011111101001000111000001000 //生成 E"后 6 行
10101000100010100000000000000 //转置 E"
10001100010010100000000000000 // 生成 V_T
10101010101010000000000000000 //转置 V
11001110111010100000000000000 //encodeu
11101101001100000000000000001 // B"=S'A+E'
10001000100010100000000000001 //C' = V+encode(u')
00100000001000000000000100000 // 吸收 B"前半段
00100000001000100000000100001 //吸收 B"后半段
00011111110100001000011100000 //吸收 C'
00011111111100000000110100000 //吸收 salt 前 192bit
01000000000000000101010000001 //执行一次吸收
00011111111101100000010100000 //吸收 salt 后 64bit
00011111111011000001000000000 //吸收 k' 128bit
01000000000000000000110000001 //执行一次吸收
01011111111110000000100000000 //挤出 ss

4.1.2. Frodo976

Keygen
11001011111000000000000000011 //吸收 5f absorb_s
00011111110000000011001000000 //吸收 seedse 长度 384
01110101001000000000000100100 // genS
01100000001000000000000000101 //genE
00011111111000000001000000010 //吸收 z absorb 长度 128
01000000000000000000100000001 //生成 seedA, absorb_ex,长度 128
01011111101000000001001000000 //生成 seedA,长度 128
11100010011001000000000000000 // B=AS+E
00011111101000000001001000010 //absorb 吸收 seedA,长度 128
00000000001000000000000000001 // pkh,吸收矩阵 B
01011111101001000001101000000 // 挤出 pkh,长度 192

Encaps
00011111101000000000100100010 //absorb 吸收 seedA,长度 128 00000000001000000000000000001 // pkh,吸收矩阵 B 01011111101001000000110100000 // 挤出 pkh,长度 192 00011111101001000000110100010 // 吸收 pkh,长度 192 00011111111001000000110000000 //吸收 u,长度 192 00011111111100000001100100000 //吸收 salt，长度 384 01000000000000000011000000001 //执行吸收，长度 768 01011111110000000001100100000 //生成 seedse，长度 384 01011111111011000000110000000 //生成 k，长度 192 11010010110000000000000000011 //吸收 96 00011111110000000001100100000 //吸收 seedse 01110101001000000000000000100 //生成 S' 01100000001000000000000101101 //生成 E' 01011111101000000001100000110//生成 E"前 6 行（384bit） 10100000000000000000000000001 //执行一次挤压 01011111101011000010100000100 //生成 E"后 10 行（640bit） 10101000100010100000000000000 //转置 E" 10001100010010100000000000000 // 生成 V_T 10101010101010000000000000000 //转置 V 11001110111010100000000000000 //encodeu 11101101001100000000000000001 // B'=S'A+E' 10001000100010100000000000001 //C = V+encode(u) 00100000001000000000000100000 // 吸收 B'前半段 00100000001000100000000100001 //吸收 B'后半段 00011111110100000000110110000 //吸收 C 前 192bit 01000000000000000100010000001 //执行一次吸收 00011111110101100011010110000 //吸收 C 后 832bit 00011111111100000001000100000 //吸收 salt 前 256bit 01000000000000000100010000001 //执行一次吸收 00011111111111000000100100000 //吸收 salt 后 128bit 00011111111011000000110000000 //吸收 k 192bit 01000000000000000001010000001 //执行一次吸收 01011111111111000000110000000 //挤出 ss,192
Decaps

```
10010111000010010000000000000 //M=C-B'S
110 0100 0100 1010 1000000000000 // u' = decode(u)
00011111101001000000110100010 //  吸收 pkh,长度 192
00011111111001000000110000000   //吸收 u,长度 192
00011111111100000001100100000   //吸收 salt，长度 384
01000000000000000011000000001   //执行吸收，长度 768
01011111110000000001100100000   //生成 seedse，长度 384
01011111111011000000110000000   //生成 k，长度 192
00100000001000000000000100000 //  吸收 B'前半段
00100000001000010000000100001 //吸收 B'后半段
00011111110100000000110110000 //吸收 C 前 192bit
01000000000000000100010000001 //执行一次吸收
00011111110101100011010110000 //吸收 C 后 832bit
00011111111110000001000100000 //吸收 salt 前 256bit
01000000000000000100010000001 //执行一次吸收
00011111111111000000100100000   //吸收 salt 后 128bit
00011111111011000000110000000 //吸收 k 192bit
01000000000000000001010000001 //执行一次吸收
010 1111 1101 0110 000011 01 00000 //挤出 ss‘,192
11010010110000000000000000011 //吸收 96
00011111110000000001100100000 //吸收 seedse
01110101001000000000000000100 //生成 S'
01100000001000000000000101101 //生成 E'
01011111010000000011000001l0//生成 E"前 3 行（384bit）
10100000000000000000000000001 //执行一次挤压
01011111101011000101000001l00 //生成 E"后 5 行（640bit）
10101000100010100000000000000 //转置 E"
10001100010010100000000000000 //  生成 V_T
10101010101010000000000000000 //转置 V
11001110111010100000000000000 //encodeu
11101101001100000000000000001 // B'=S'A+E'
10001000100010100000000000001 //C = V+encode(u)
00100000001000000000000100000 //  吸收 B'前半段
00100000001000010000000100001 //吸收 B'后半段
00011111110100000000110110000 //吸收 C 前 192bit
```

0100000000000000100010000001 //执行一次吸收
0001111111010110011010110000 //吸收 C 后 896bit
0001111111110000000110100000 //吸收 salt 前 192bit
0100000000000000100010000001 //执行一次吸收
0001111111110110000110100000 //吸收 salt 后 192bit
0001111111101100000110000000 //吸收 k 192bit
0100000000000000000110000001 //执行一次吸收
010 1111 1111 1100 000011 0000000 //挤出 ss',192

4.1.3. Frodo1344

Keygen
11001011111000000000000000011 //吸收 5f absorb_s
0001111111000000010000100000 //吸收 seedse 长度 512
01110101001000000000000100100 // genS
01100000001000000000000000101 //genE
0001111111100000000100000010 //吸收 z absorb 长度 128
0100000000000000000100000001 //生成 seedA, absorb_ex,长度 128
0101111110100000000100100000 //生成 seedA,长度 128
11100010011001000000000000000 // B=AS+E
0001111110100000000100100010 //absorb 吸收 seedA,长度 128
0000000000100000000000000001 // pkh,吸收矩阵 B
0101111110100100001000100000 // 挤出 pkh,长度 256
Encaps
0001111110100000000100100010 //absorb 吸收 seedA,长度 128
0000000000100000000000000001 // pkh,吸收矩阵 B
0101111110100100001000100000 // 挤出 pkh,长度 256
0001111110100100001000100010 // 吸收 pkh,长度 256
0001111111100100001000000000 //吸收 u,长度 256
0001111111110000010000100000 //吸收 salt，长度 512
0100000000000000100000000001 //执行吸收，长度 1024
0101111111000000010000100000 //生成 seedse，长度 512
0101111111101100001000000000 //生成 k，长度 256
11010010110000000000000000011 //吸收 96
0001111111000000010000100000 //吸收 seedse,长度 512
01110101001000000000000000100 //生成 S'

01100000001000000000000101101 //生成 E'
01011111101000000110100001l0//生成 E"前 13 行（832bit）
10100000000000000000000000001 //执行一次挤压
01011111101110100001100001l00 //生成 E"后 3 行（640bit）
10101000100010100000000000000 //转置 E"
10001100010010100000000000000 // 生成 V_T
10101010101010000000000000000 //转置 V
11001110111010100000000000000 //encodeu
11101101001100000000000000001 // B'=S'A+E'
10001000100010100000000000001 //C = V+encode(u)
00100000001000000000000100000 // 吸收 B'前半段
00100000001000100000000100001 //吸收 B'后半段
00011111110100000111101100000 //吸收 C 前 960bit
01000000000000000100010000001 //执行一次吸收
00011111111101110000010110000 //吸收 C 后 64bit
00011111111110000010000100000 //吸收 salt 前 512bit
00011111111011000010000000000 //吸收 k 256bit
01000000000000000011010000001 //执行一次吸收,832
01011111111110000010000000000 //挤出 ss,256
Decaps
10010111000010010000000000000 //M=C-B'S
110 0100 0100 1010 1000000000000 // u' = decode(u)
00011111101001000010000100010 // 吸收 pkh,长度 256
00011111111001000010000000000 //吸收 u,长度 256
00011111111110000010000100000 //吸收 salt，长度 512
01000000000000000100000000001 //执行吸收，长度 1024
01011111110000000010000100000 //生成 seedse，长度 512
01011111111011000010000000000 //生成 k，长度 256
00100000001000000000000100000 // 吸收 B'前半段
00100000001000100000000100001 //吸收 B'后半段
00011111110100000111101100000 //吸收 C 前 960bit
01000000000000000100010000001 //执行一次吸收
00011111111011100000101100000 //吸收 C 后 64bit
00011111111110000010000100000 //吸收 salt 前 512bit
00011111111011000010000000000 //吸收 k 256bit

0100000000000000011010000001 //执行一次吸收,832
010 1111 1101 0110 000100 0100000 //挤出 ss,256
11010010110000000000000000011 //吸收 96
00011111110000000100001000000 //吸收 seedse,长度 512
01110101001000000000000000100 //生成 S'
01100000001000000000000101101 //生成 E'
01011111101000000110100001l0//生成 E"前 13 行（832bit）
10100000000000000000000000001 //执行一次挤压
01011111101110100001100001000 //生成 E"后 3 行（640bit）
10101000100010100000000000000 //转置 E"
10001100010010100000000000000 // 生成 V_T
10101010101010000000000000000 //转置 V
11001110111010100000000000000 //encodeu
11101101001100000000000000001 // B'=S'A+E'
10001000100010100000000000001 //C = V+encode(u)
00100000001000000000000100000 // 吸收 B'前半段
00100000001000100000000100001 //吸收 B'后半段
00011111110100000111101100000 //吸收 C 前 960bit
01000000000000000100010000001 //执行一次吸收
00011111111011100000101100000 //吸收 C 后 64bit
00011111111100000100001000000 //吸收 salt 前 512bit
00011111111011000010000000000 //吸收 k 256bit
01000000000000000011010000001 //执行一次吸收,832
01011111111110000010000000000 //挤出 ss,256

4.2. 测试数据集

4.2.1. Frodo640

Keygen input (s, $seed_{SE}$, z)

//s
7c9935a0b07694aa
0c6d10e4db6b1add
//seedSE
2fd81a25ccb14803
2dcd739936737f2d
b505d7cfad1b4974
99323c8686325e47
//z
92f267aafa3f87ca
60d01cb54f29202a

Keygen output (pkh, pk=$seed_A$ + b)
//pkh
13ca7d97b3852edb
48e21b7c5088fbad
//$seed_A$(pk)
a3b0d78801479de0
f67b9cd5bcca3d43
//b(pk)
dca7fe2c1a374ad2, 9e5b3a7278dcda00, 369aeb448afff61c, 7fe2d06fcfede30f,
a33ddf288e9e5df3, 1c04dd1b2a46d8f8, 503112fecb4acaf1, c5ca23109487786c,
fd219af01748b394, 57077cdbdeeaaf6f, 08af4e409415ffc7, 49e6be6fd4a30cdc,
ce7a594e9ccd9359, a315855227d3a3c7, 5c17ec427b96b3ea, 35b98faedc69ffa3,
8ab0f110613dac13, bcc42a28864b2f60, 1613797ad0c15df9, f3b691a7e9899738,
cd23cad2634936e9, d46e23cd5fa86bd2, efc4d5d63da0ea78, 195c09aaf3f9d972,
2d6aa17009990534, b180da8e5e243f3d, 07607a7b6137013b, 889f4807f803a740,
530763e355d472d2, a48e4e5f752aae66, 42aa859f6ccd75c4, 91f6413cd16f0c4b,
ba47de73158c2358, 3fdc7c294b308fac, 0b74ae831439a2d1, 7eb4208271ba3928,
8423184030edd555, d698d0c76d544e41, b9dad00dee02db34, f1363cfd5656c428,

b2edc9ca4ffd9a6d, dc1a88cff99cdad3, 88d8258d39934441, a405e05363376493,
889e09dbae8ad2bf, 62bdf6ac9a3da3fb, e90714786c6b6f69, 4be4eafe724f692c,
0469d0a92c84e317, af658fbdfca6da0b, bf794e63e21dac91, b3712b2746e957c0,
93c8d2cfee97e83c, 89e160f82417a8e9, 507e9f3d0e43be44, 08f8dc404e23adf3,
8ada0e2a020fbf39, dcce655f9a13b648, cdc13c41f54157bf, 055fe1bc15223713,
e64224d49b953f19, cbf763e6985c8885, 5a80ec73f5f19ed7, 166de003ba7bd8e6,
f97c3e996dd9b38f, 847ab7dccd3b4fe5, 92641e534329af1f, 61ae4930680f66e2,
635e5205e84632db, c44a89d0c9d531f6, 256908787de9748e, 34dbb85809c40ac6,
78621889a720deb4, 6f304e5713ecfe47, 3519ce637deadee8, 27956a8b878f3e40,
40e588db6830ada2, d6c05e59546287f0, 2a57f8c51e24ef3b, b8c096c2b88dd15e,
f65a4da05e83cd5a, 08307d98b2745855, dcfd0d7bf99342f4, 48d0af9f922e276e,
ca05f002569c33a5, 3167aa3889f6fce9, e54ac997154426b6, b0a5051c2cba66cc,
50f08353a997a5b6, a44049a8c885d59b, e0b53d662086896f, 48af68cbb8dcb123,
e09c1346b3bd8900, b34e66ae87ea62db, 03da83a9d2f71ca7, d00ecc01fce8646e,
69e1e2db022bc447, 951eb3159ace3f3e, 7f097848556257e8, a5c093f2eff7788c,
2c084a610a626754, 8163da04278435d1, c3c99c769eebf934, b255e61eb393f2de,
b51756bffccc3550, d12c76435f42d2a4, 9e3218bd0b6a2380, a867d96123ac0fe3,
3d7bb78b5a22087f, 573bbb3e8605ae00, 1febdb4bc45e2d52, f1c5eedd77bddefe,
f590761ef4958c03, a527b2bfabaab3d8, b34d661bee2f9f1d, c916db821767e61f,
1ac779f4d62d089c, 5be098f6f5fbdd2e, cee2c146246e426b, b6f62423d6962ba4,
5835e006a179766f, 5b50e6b10dfe831d, 5d17c62d3e899b82, 930501a817f79355,
b5ba9c02f8a911e0, 48ce5872417d72b5, a9a93bb746d05b9c, 0561b2b0c86326c6,
13ce135b678f812d, da125941696ccde5, 26a9fe7401cda8b2, c28b23cd3eb2c2e4,
e06591fcf2dab373, 3b5555946f6b5ecd, f73b7904468b70c9, af9861bf821e0859,
ed6bb8dcc4ce7dcd, 5b9655611e11cda4, 73288367d9284518, a1d6a13f7539a547,
391b4b401ec86823, c069d427fae24b61, 6f7995a008decca6, af98def67ec0a527,
db02188dd3e3d797, 11c141962035d9a8, 3d5953acc7ad37bb, de2a4e0cd5e4c7e2,
13bc3c78aaf38cae, 88e996db99786420, 5830c6d149e034cb, 5d1366684d5239ea,
7814e8ef34b6b6e9, b6487a9964051401, 05bc1b4a2094eb7c, e0839cc44838c363,

56693ffba147225a, 841c38f2b9a6e166, 19dfc443b1271836, 298ccdfd9d31ddff,
3a9c822fe09c0815, 3564a5e1957617d8, 590b691899f879fc, 170e041eff84b835,
657b0bed88b509a7, cea1e70e35b59276, f9857ba8c309817d, 1435e663dead99a7,
a16f8c92d48cacaf, 064e1c4eee240b4a, f350e500b86feef9, 7ff1994df95195aa,
a906d751fa4ed247, 776d5e56e29ffdf4, 6073f36c442a7bac, 17803955bc36694a,
bb3b279473e8efab, 94a17735cbe3466f, 85b695601867745e, 36cddc4fa224d7ee,
1f5b7e7a8886c6e0, 1b10a28798619086, 258b618cca2c2ad1, b4f47acb107dc1c7,
efceafd84fb7cdbd, 5f33a783ebe7e16e, 1001e0fc9daa0a44, 54a41d418cfd5c2b,
9b38cbe2a5708834, bbd763169341869e, 603a29f484e46489, a2f803c8a82fe9d6,
5fbd61ba680a899f, 3bf8f16df838cac4, 542581049d5e684a, 82033024c74a5d19,
3e707069f20507d9, 172c8a1823bc236a, 225782c43b4b9744, 3dcdf6cfd26cfec3,
90877223ce407584, 121ba4e92755aaf9, 94343a89a1f615da, fda32b7b358248fb,
31032a0d65512389, bc26f07cb272b8e8, d5c7ee59e4585e75, 17b58d509ce0f168,
15d3fb9aeb301a48, e79d643cce7d6c40, c3eee1b5dc871641, 74c36e30baada505,
314f8767c9237fad, 70f5ef54d8b0c93b, af05a6006083e260, a906af281462603e,
2cec281f2cf45e58, c77004576aac323d, bc94774dcb16de7d, 146e985c42f7a244,
607b32db765b415f, 12fd789a6a1e7359, b3fb73031410af54, 2c1b14c2d00e3f43,
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Encaps input (pk, u, salt)

//pk 见 Keygen output
//u
eb4a7c66ef4eba2d
db38c88d8bc706b1
//salt
d639002198172a7b
1942eca8f6c001ba
26202bee59ac2754
84ea767d41d8d357

Encaps output (ss, ct=c1+c2+salt)

//ss
2ed42ce7d5dbfb11
5f2e2bdcb650b3fa
//c1(ct)
1c35c98b0605557f, e74d1dd13d09c275, a605768436b364fc, 2b17968701fecdf4,
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4ac46e6f7a98ac0f,  b53b2e4e9c2fe498,  ab4bc8a902c0bd8f,  28d8af850e65b62b,
//c2(ct)
12441bd2bb605d73,  7e872be0ca7c759f,  87e7188de87c98bf,  22065fb4f4f11f55,
47f17e5a78e15644,  b40e90aa477e45a4,  a1109f0394f219b0,  1c28fc762b390be6,
e20b8f33dda25567,  1e8fa01a187e6f66,  b897e9e6296d3046,  73379cf5d730d09d,
```

f7e3efe77c23f530, 61de06b22a9f36b4, 4990e68872f88590, //salt 见 Encaps input
Decaps input (ct)
ct 见 Encaps output
Decaps output (ss)
//ss 2ed42ce7d5dbfb11 5f2e2bdcb650b3fa

4.2.2. Frodo976

Keygen input（s，seedSE，z）

//s
7c9935a0b07694aa
0c6d10e4db6b1add
2fd81a25ccb14803
//$seed_{SE}$
2dcd739936737f2d
b505d7cfad1b4974
99323c8686325e47
92f267aafa3f87ca
60d01cb54f29202a
3e784ccb7ebcdcfd
//z
45542b7f6af77874
2e0f4479175084aa

Keygen output (pkh, $pk=seed_A+b$)
//pkh 9b1f7a8d8a66043e da842896a37250af 2c80a312dd63c7cf //$seed_A$(pk) 01fdc97c2d10877b 667820a9462536ae

```
//b(pk)
18a16776fb224755,  5b3ae80175e9fd16,  76c72950d185a26b,  6f45aff7f5b6a1fc,
21220079605491a6,  4586e8457f3bec89,  2886461e805ac509,  91950c7c910f1e7c,
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Encaps input（pk，u，salt）			

pk 见 Keygen output
//u
ee716762c15e3b72
aa7650a63b9a5100
40b03c0fe70475c0
//salt
463bbc45a0ba5b79
80dd46eef82fb062
035077d042f306bb
6391040e0dd965f1
fda9d183ca9fccb4
8fc010b184ab0033

Encaps output（ss，ct=c1+c2+salt）

//ss
5b6e5a69a3d5f8e7
5eea3a6e95595ed0
278da55b8b373142
//c1(ct)
5effc1c7a68772de, 82f3b2d91b1b2151, 39f6930b9b99a8ae, 3a18c17f65ac63a0,
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03af65d1780f021e, d1234fdda7e78017, 76d5f9e40a59eae9, ce6fe124aff9a8af,
//salt 见 Encaps input

Decaps input(ct)
//ct 见 Encaps output
Decaps output(ss)
//ss 5b6e5a69a3d5f8e7 5eea3a6e95595ed0 278da55b8b373142

4.2.3. Frodo1344

Keygen input（s，$seed_{SE}$，z）
//s 7c9935a0b07694aa 0c6d10e4db6b1add

2fd81a25ccb14803
2dcd739936737f2d
//$seed_{SE}$
b505d7cfad1b4974
99323c8686325e47
92f267aafa3f87ca
60d01cb54f29202a
3e784ccb7ebcdcfd
45542b7f6af77874
2e0f4479175084aa
488b3b74340678aa
//z
38e22e9628b0a161
fdeb0bd252173b9c

Keygen output（pkh，pk=$seed_A$+b）

//pkh
3a3673875d692016
275e7b0bf922ba96
b1a72312d54504df
38d47635e5677d79
//$seed_A$
44ff290750370216
f0c737c42963c9cf
//b(pk)
672fd1cdf473c6b1, 84ab57df33d6b786, c6d7a7259e25d105, 0107f83c6c6cb77c,
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Encaps input（pk，u，salt）

//pk 见 Keygen output
//u
9f08587687ff6676

5c671de73e918d28 23ca573ff4e7a31a 9160324026e540ea //salt cb3a04e0d54c75de b9705bfdfbdf935a 7528802ee6e5b0c6 a73b2b761d9bd084 8a6e4cf3fc4ca84f 14e0331af35bfef4 1e42b13a6dae6df9 37f738c1857ba1ca
Encaps output（ss，ct=c1+c2+salt）

//ss
8d20f971464df19e
0561bdd385afd0e2
ef0ce212efd45a63
2f5d2c64f3d66aac
//c1
e615bb53c184fa82, 8ce2ebac96ef23e9, eddb35945bdd06f9, 7cc93e94fa93ca3c,
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//salt 见 Encaps input			
Decaps input（ct）			
ct 见 Encaps outpu			
Decaps output（ss）			

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// ss
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